

VERAVAL, India

WMO: 429090

Lat: 20.900N

Lon: 70.367E

Elev: 8

StdP: 101.23

Time Zone: 5.50 (IND)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	15.0	16.2	-1.8	3.2	20.4	1.2	4.1	20.9	7.9	24.6	7.2	24.8	2.6	50	0.398

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
6	4.0	34.9	23.6	33.9	25.5	33.1	26.3	29.3	32.2	28.9	31.8	28.6	31.4	3.3	50

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
28.5	24.9	31.4	28.1	24.3	31.0	27.7	23.8	30.6	96.6	32.4	94.5	31.9	92.7	31.5	31.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
11.1	9.4	8.2	DB	12.2	38.9	1.4	1.1	11.2	39.7	10.3	40.3	9.5	40.9	8.5	41.7	
			WB	7.0	29.8	1.7	0.7	5.8	30.3	4.8	30.7	3.8	31.1	2.6	31.6	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
Temperatures, Degree-Days and Degree-Hours	DBAvg	27.2	22.7	23.7	26.1	28.1	29.9	30.5	28.9	28.0	28.3	29.1	27.2	24.3	
	DBStd	2.79	1.57	1.91	1.79	1.42	1.00	1.15	1.20	0.99	1.03	1.22	1.47	1.73	
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0	
	HDD18.3	1	0	1	0	0	0	0	0	0	0	0	0	0	
	CDD10.0	6295	392	382	500	542	616	616	587	558	549	592	517	444	
	CDD18.3	3254	134	149	242	292	358	366	329	299	299	333	267	186	
	CDH23.3	35604	939	1068	2054	3183	4644	4963	4039	3291	3383	3912	2652	1476	
	CDH26.7	14072	199	271	687	1173	2192	2570	1584	888	1108	1761	1155	484	
Wind	WSAvg	3.7	2.8	3.0	3.3	3.5	4.2	5.3	6.4	5.3	3.4	2.4	2.3	2.5	
Precipitation	PrecAvg	819	1	1	0	0	2	171	335	174	91	18	8	0	
	PrecMax	1888	10	20	12	11	39	929	1532	640	318	172	163	9	
	PrecMin	167	0	0	0	0	0	1	17	11	0	0	0	0	
	PrecStd	401	2	3	2	2	6	164	241	161	88	32	26	1	
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	31.2	32.6	35.5	36.6	34.1	34.5	32.6	31.6	33.0	36.8	34.9	32.6	
		MCWB	19.1	19.3	20.4	21.8	28.3	28.7	28.4	27.4	27.6	22.5	21.6	21.2	
	2%	DB	29.7	30.7	32.8	33.0	33.2	33.8	31.9	30.7	31.9	35.1	33.7	31.4	
		MCWB	19.1	20.3	21.7	24.9	28.2	28.7	28.0	27.2	27.3	23.8	22.4	21.1	
	5%	DB	28.7	29.4	31.1	31.7	32.8	33.1	31.2	30.1	31.3	34.0	32.8	30.5	
		MCWB	19.4	20.8	23.1	26.2	28.1	28.5	27.8	26.9	26.9	25.0	22.8	21.0	
	10%	DB	27.7	28.3	29.9	31.0	32.2	32.6	30.6	29.5	30.7	32.8	31.8	29.5	
		MCWB	19.5	21.0	23.5	26.2	27.8	28.4	27.5	26.8	26.7	25.7	22.9	20.8	
	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	24.6	25.3	27.0	28.5	29.6	29.9	29.2	28.2	29.1	29.2	28.1	25.6
			MCDB	28.1	28.8	30.5	31.3	32.6	32.8	31.4	30.1	31.5	32.8	31.4	29.8
		2%	WB	23.3	24.0	26.0	27.8	29.1	29.4	28.6	27.7	28.2	28.3	26.8	24.4
			MCDB	27.4	28.1	29.6	30.7	32.3	32.4	30.8	29.6	30.7	31.9	30.7	28.9
5%		WB	22.2	23.2	25.2	27.2	28.6	29.1	28.2	27.4	27.6	27.7	25.9	23.5	
		MCDB	26.6	27.4	28.9	30.3	31.8	32.1	30.3	29.2	30.1	31.4	30.1	28.3	
10%		WB	21.2	22.4	24.6	26.7	28.2	28.7	27.8	27.1	27.2	27.1	25.0	22.6	
		MCDB	26.1	26.7	28.3	29.8	31.3	31.6	29.9	28.8	29.5	30.7	29.6	27.6	
Mean Daily Temperature Range		5% DB	MDBR	11.3	10.2	9.0	7.1	5.3	4.0	2.7	2.7	4.6	8.4	10.3	11.0
			MCDBR	11.8	11.4	10.7	8.2	5.5	4.8	3.5	3.5	5.7	10.4	11.0	11.5
	MCWBR		8.5	8.4	7.8	4.9	2.4	2.1	1.9	1.8	2.8	6.6	7.4	8.0	
	5% WB	MCDBR	10.4	9.6	8.4	6.7	5.3	4.5	3.2	3.0	4.9	7.9	9.4	10.4	
		MCWBR	8.9	7.9	6.3	4.0	2.6	2.3	2.0	1.9	2.9	5.2	7.2	8.4	
Clear-Sky Solar Irradiance	taub	0.406	0.410	0.463	0.498	0.549	0.699	0.740	0.667	0.570	0.540	0.516	0.442		
	taud	2.250	2.250	2.108	2.049	1.971	1.692	1.628	1.767	1.972	1.971	1.993	2.144		
	Ebn at Noon	859	881	846	819	771	658	632	682	746	748	741	804		
	Edh at Noon	127	135	161	173	186	243	260	227	182	176	164	138		
All-Sky Solar Radiation	RadAvg	4.90	5.78	6.74	7.29	7.36	5.64	3.98	4.18	5.20	5.59	4.89	4.59		
	RadStd	0.14	0.16	0.13	0.13	0.24	0.50	0.46	0.42	0.44	0.24	0.23	0.21		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (1 neighbor)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page